# Health-Aware High-Frequency Trading for Battery Energy Storage Systems: A Multi-Objective Dynamic Programming Approach

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## Abstract

Continuous intraday electricity markets operate at a millisecond pace, where liquidity and price volatility create significant arbitrage opportunities for Grid-scale Battery Energy Storage Systems (BESS). However, traditional optimization methods like Mixed-Integer Linear Programming (MILP) are computationally too expensive to react to every relevant update in the Limit Order Book (LOB). Furthermore, existing high-frequency trading strategies often neglect the non-linear degradation characteristics of battery cells, leading to suboptimal lifecycle economics. This paper presents a novel health-aware high-frequency trading framework that integrates a fast Dynamic Programming (DP) approximation of the Rolling Intrinsic (RI) policy with a sophisticated battery degradation model. Our approach incorporates cycle depth-dependent aging costs through a piecewise linear approximation, enabling realistic modeling of electrochemical battery degradation while maintaining computational efficiency. We demonstrate that our DP formulation achieves solution speeds 2-3 orders of magnitude faster than exact MILP solutions while preserving solution quality. Backtesting on German intraday market data from 2021 shows that our health-aware policy not only maximizes short-term trading profits but also extends battery lifespan by intelligently managing cycle depth and frequency. The proposed framework enables truly high-frequency trading—reacting to every relevant LOB update—while ensuring long-term asset value preservation.

**Keywords:** Battery Energy Storage, High-Frequency Trading, Dynamic Programming, Battery Degradation, Intraday Markets, Cycle Aging, Limit Order Book

## 1 Introduction

The rapid growth of renewable energy sources has introduced significant variability into electrical grids worldwide. Battery Energy Storage Systems (BESS) have emerged as a critical solution for managing this variability, providing flexibility through energy arbitrage, frequency regulation, and demand response services [1, 2]. In liberalized electricity markets, the economic viability of BESS depends critically on their ability to participate profitably in various short-term energy markets, particularly the continuous intraday (ID) market.

### 1.1 Motivation and Background

Continuous intraday electricity markets in Europe, particularly in Germany, operate at a millisecond pace with trading continuing until 5 minutes before delivery. This high-frequency environment creates numerous arbitrage opportunities as market participants continuously update their positions based on evolving forecasts and market conditions. The German continuous intraday market recorded over 240 million order submissions in 2021—averaging 7.6 orders per second—underscoring the necessity of algorithms that can process and respond to market updates in real-time.

However, two fundamental challenges limit the practical deployment of BESS in high-frequency trading:

**Computational Challenge:** Traditional optimization methods such as MILP are too slow to evaluate trading decisions at the pace required by continuous markets. When the market updates every millisecond, strategies that require seconds or minutes to optimize become obsolete before execution.

**Battery Health Challenge:** Electrochemical batteries experience accelerated degradation when

subjected to frequent charge-discharge cycles, particularly at deep discharge levels. The relationship between cycle depth and battery aging is highly non-linear—a battery cycled at 10% depth of discharge (DoD) can perform over 50,000 cycles, while the same battery cycled at 100% DoD may only survive 500 cycles [5]. Current high-frequency trading strategies typically use oversimplified linear degradation models or ignore degradation entirely, leading to strategies that maximize short-term profits at the expense of long-term asset value.

### 1.2 Research Gap

Existing literature on BESS trading strategies can be broadly categorized into two streams:

**Market-focused approaches** [3, 4] optimize trading decisions across multiple markets but typically operate at hourly or 15-minute intervals, missing the high-frequency arbitrage opportunities available in continuous markets. These approaches are computationally tractable but fail to capture the full revenue potential of fast-moving markets.

**Degradation-aware approaches** [5, 6] incorporate sophisticated battery aging models but use static optimization horizons and do not address the computational challenges of high-frequency decision-making. The rolling intrinsic policy has been shown to be nearly optimal for gas storage [7] and naturally extends to batteries, but its application to high-frequency continuous markets with realistic degradation models remains unexplored.

Recent work by Semmelmann et al. [8] and Schaurecker et al. [9] has begun to address high-frequency trading for BESS on continuous intraday markets. However, these approaches either neglect degradation costs entirely or use oversimplified linear models that do not capture the true non-linear aging characteristics of electrochemical batteries.

### 1.3 Contributions

This paper bridges the gap between computationally efficient high-frequency trading and realistic battery health management by making the following contributions:

1. We formulate a health-aware Rolling Intrinsic (RI) policy that integrates the piecewise linear cycle aging cost model of Xu et al. [5] with the high-frequency trading framework of Schaurecker et al. [9].

2. We develop a Dynamic Programming (DP) approximation that solves the intrinsic optimization problem 2-3 orders of magnitude faster than exact MILP formulations, enabling reaction to every relevant Limit Order Book (LOB) update while accounting for non-linear degradation.

3. We incorporate depth-of-discharge (DoD) dependent aging costs through a multi-segment piecewise linearization that provides an accurate approximation of electrochemical battery cycle aging while remaining compatible with fast DP solution methods.

4. We demonstrate through extensive backtesting on 2021 German intraday market data that our health-aware policy achieves superior lifecycle economics compared to degradation-agnostic strategies, extending battery lifespan while maintaining competitive trading profits.

5. We provide a comprehensive analysis of the trade-offs between trading frequency, degradation costs, and battery parameters, offering insights for both operators and market designers.

### 1.4 Paper Organization

The remainder of this paper is organized as follows: Section 2 reviews the continuous intraday electricity market structure and the limit order book mechanism. Section 3 presents the battery degradation model, including the rainflow cycle counting method and piecewise linear cost approximation. Section 4 develops the health-aware rolling intrinsic policy and its efficient DP formulation. Section 5 describes our backtesting methodology and presents numerical results from the German market. Section 6 discusses implications and extensions, and Section 7 concludes.

## 2 Continuous Intraday Electricity Markets

### 2.1 Market Structure

European electricity markets operate primarily through two exchanges: NordPool (Scandinavian region) and EPEX (Continental Europe). The Single Intraday Coupling (SIDC) initiative has unified these markets, enabling cross-border trading subject to transmission capacity constraints. This paper focuses on the German market, which is Europe's largest and trades primarily through EPEX SPOT.

The cascade of electricity markets ranges from long-term forward contracts to day-ahead auctions to

continuous intraday trading. The continuous intraday market serves as the final opportunity for market participants to adjust positions before delivery, enabling them to respond to updated forecasts and minimize imbalance charges from the transmission system operator (TSO).

In Germany, the continuous intraday market opens at 15:00 on the day before delivery, allowing up to 32 hourly products to trade simultaneously. Trading for SIDC products halts 60 minutes before delivery, while German-only products continue trading until 5 minutes before delivery.

## 2.2 Limit Order Book Mechanics

Continuous markets operate through a limit order book (LOB) mechanism. Each product $t$ (representing a delivery hour) has an order book $\mathcal{O}_t = \mathcal{O}_t^+ \cup \mathcal{O}_t^-$ where:

- $\mathcal{O}_t^+$ contains all active ask (sell) orders
- $\mathcal{O}_t^-$ contains all active bid (buy) orders

Each order $i \in \mathcal{O}_t$ is characterized by:

- Limit price $P_i$ (€/MWh)
- Quantity $Q_i$ (MWh)
- Validity period
- Order type (buy/sell)

When a new limit order arrives, it is immediately matched against existing orders if possible. Orders are matched according to price-time priority: the best-priced orders are matched first, and among orders at the same price, earlier submissions have priority.

The bid-ask spread $\delta_{t^*,t}$ at time $t^*$ for product $t$ is defined as:

$$\delta_{t^*,t} = \min_{i \in \mathcal{O}_t^+} P_i - \max_{j \in \mathcal{O}_t^-} P_j \tag{1}$$

A key challenge in continuous markets is the lack of liquidity far from delivery. Figure 2 in Schaurecker et al.[9] shows that while thousands of orders are posted hours before delivery, this drops to hundreds for products 10+ hours away and to mere handfuls at the start of trading. This creates wide bid-ask spreads and high price volatility for products distant from delivery.

# 3 Battery Degradation Modeling

## 3.1 Electrochemical Battery Degradation Mechanisms

Electrochemical batteries have limited cycle life due to fading of active materials caused by charge-discharge cycles. This degradation process is similar to fatigue in materials subjected to cyclic mechanical loading [5]. For lithium-ion batteries, which dominate grid-scale applications, degradation is influenced by several stress factors that can be categorized as:

**Non-operational factors:**

- Ambient temperature and humidity
- Battery state of life
- Calendar aging

**Operational factors:**

- Cycle depth (DoD)
- Over-charge and over-discharge
- Current rate (C-rate)
- Average state of charge (SoC)

For market dispatch optimization, cycle depth is the most critical factor. Laboratory tests on 7 Wh Lithium Nickel Manganese Cobalt Oxide (NMC) cells show dramatic differences in lifetime energy throughput based on cycle depth:

- At 10% DoD: $> 50,000$ cycles, yielding 35 kWh lifetime throughput
- At 100% DoD: $\approx 500$ cycles, yielding only 3.5 kWh lifetime throughput

This ten-fold difference in lifetime energy throughput demonstrates the critical importance of accounting for cycle depth in optimization models.

## 3.2 The Rainflow Cycle Counting Algorithm

The rainflow counting algorithm, widely used in materials stress analysis, provides a method to identify and quantify cycles from a time-series SoC profile. Given a SoC profile with local extrema $s_0, s_1, s_2, \ldots$, the algorithm identifies cycles as follows:

The rainflow residue contains only half cycles linking adjacent local extrema. A half cycle with decreasing SoC is a discharging half cycle, while increasing SoC indicates a charging half cycle.

**Algorithm 1** Rainflow Cycle Counting

Start from the beginning of the SoC profile
Calculate $\Delta s_1 = |s_0 - s_1|$, $\Delta s_2 = |s_1 - s_2|$, $\Delta s_3 = |s_2 - s_3|$
**if** $\Delta s_2 \leq \Delta s_1$ AND $\Delta s_2 \leq \Delta s_3$ **then**
  A full cycle of depth $\Delta s_2$ has been identified
  Remove $s_1$ and $s_2$ from profile
  Repeat using points $s_0, s_3, s_4, \ldots$
**else**
  Shift forward and repeat using $s_1, s_2, s_3, s_4, \ldots$
**end if**
Repeat until no more full cycles can be identified

Remaining profile forms the rainflow residue =0

### 3.3 Cycle Depth Stress Function

The total life loss $L$ from a SoC profile is the sum of life loss from all cycles identified by the rainflow algorithm:

$$L = \sum_{i=1}^{I} \Phi(\delta_i) \tag{2}$$

where $I$ is the number of cycles and $\Phi(\delta)$ is the cycle depth stress function.

For Li(NiMnCo)$O_2$-based lithium-ion cells, experimental data yields a near-quadratic stress function:

$$\Phi(\delta) = (5.24 \times 10^{-4})\delta^{2.03} \tag{3}$$

where $\delta \in [0, 1]$ represents the cycle depth as a fraction of total capacity.

### 3.4 Piecewise Linear Cost Approximation

While the rainflow algorithm provides an accurate ex-post assessment of battery degradation, it cannot be directly incorporated into optimization because it lacks an analytical mathematical expression. To enable optimization while preserving accuracy, we construct a piecewise linear upper-approximation of the cycle depth stress function.

We assume that battery cycle aging occurs primarily during discharge, so that a discharging half cycle causes the same aging as a full cycle of equal depth, while charging half cycles cause negligible aging. This assumption is reasonable since charged and discharged energy are nearly equal over daily timescales.

During a cycle from starting SoC $e^{up}$ to ending SoC $e^{dn}$, the cycle depth is:

$$\delta_t = \frac{e^{up} - e^{dn}}{E_{rate}} \tag{4}$$

where $E_{rate}$ is the energy capacity of the BESS (MWh).

For a battery discharged at power $g_t$ over time interval of duration $M$ (hours), the cycle depth evolves as:

$$\delta_t = \frac{1}{\eta^{dis} E_{rate}} M g_t + \delta_{t-1} \tag{5}$$

where $\eta^{dis}$ is the discharge efficiency.

The marginal cycle aging cost is obtained by taking the derivative:

$$\frac{\partial \Phi(\delta_t)}{\partial g_t} = \frac{d\Phi(\delta_t)}{d\delta_t} \cdot \frac{\partial \delta_t}{\partial g_t} = \frac{M}{\eta^{dis} E_{rate}} \frac{d\Phi(\delta_t)}{d\delta_t} \tag{6}$$

We define a piecewise linear cost function $c(\delta_t)$ with $J$ segments that evenly divide the cycle depth range $[0, 1]$:

$$c(\delta_t) = \begin{cases} c_1 & \text{if } \delta_t \in [0, \frac{1}{J}) \\ c_2 & \text{if } \delta_t \in [\frac{1}{J}, \frac{2}{J}) \\ \vdots & \\ c_j & \text{if } \delta_t \in [\frac{j-1}{J}, \frac{j}{J}) \\ \vdots & \\ c_J & \text{if } \delta_t \in [\frac{J-1}{J}, 1] \end{cases} \tag{7}$$

The marginal cost of segment $j$ is calculated by prorating the battery replacement cost $R$ (€) to the marginal aging:

$$c_j = \frac{RMJ}{\eta^{dis} E_{rate}} \left[ \Phi\left(\frac{j}{J}\right) - \Phi\left(\frac{j-1}{J}\right) \right] \tag{8}$$

This piecewise linear approximation provides an upper bound on the true cycle aging cost and approaches the exact cost as the number of segments $J$ increases. The approximation error $\epsilon$ relative to the rainflow-based benchmark is:

$$\epsilon = \frac{|\hat{C} - RL|}{RL} \tag{9}$$

where $\hat{C}$ is the predicted cost from the piecewise model and $RL$ is the benchmark cost from rainflow counting.

## 4 Health-Aware RI Policy

### Parameters

- $\mathcal{T}$: Set of time intervals in the optimization horizon.
- $\mathcal{O}_t^+$ contains all active ask (sell) orders.
- $\mathcal{O}_t^-$ contains all active bid (buy) orders.

- $\mathcal{O}_t$: Limit Order Book for product $t$ containing orders $i$.
- $P_i, Q_i$: Price and quantity of limit order $i$.
- $\nu = \nu_{trade}$: Cost of Trading
- $q_i$: Accepted quantity for order $i$.
- $M$: Duration of a market dispatch time interval.
- $J$: Number of segments in the cycle aging cost function.
- $c_j$: Marginal aging cost of cycle depth segment $j$.
- $e_{t,j}$: Energy stored in cycle depth segment $j$ at time $t$.
- $u$: Minimum Trading Unit
- $f_t^+, f_t^-$: Total buying (+) and selling (-) power.
- $f_t$: Net power exchanged with the grid.
- $f_t^0$: Previously Committed Power.
- $\bar{f}, \underline{f}$: Maximum injection (charging) and withdrawal (discharging) power limits.
- $i_t, w_t$: Total charging and discharging power at time $t$.
- $s_t$: State of Charge (energy) at time $t$.
- $\eta^+, \eta^-$: Charging and discharging efficiencies.
- $E^{min}$: Minimum energy storage capacity.
- $E^{max}$: Maximum energy storage capacity.
- $\bar{e}_j$: Maximum energy capacity of cycle depth segment $j$.
- $e_j^0$: Initial amount of energy of cycle depth segment $j$.
- $E^{final}$: Amount of energy that must be stored at the end of the optimization horizon
- $R$: Battery cell replacement cost to marginal cycle aging
- $E^{rate}$: Energy capacity of the BES
- $E^0$: Initial amount of energy stored in the BES.

## 4.1 The Classical Intrinsic Problem

The intrinsic strategy maximizes rewards from a storage system in a perfectly liquid futures market. For a set of time periods $\mathcal{T} = \{1, 2, \ldots, T\}$ with futures prices $P_t$, the classical intrinsic problem optimizes traded quantities $q_t$ as:

$$\max_{q_t} \quad \sum_{t=1}^{T} P_t(-q_t) \tag{10a}$$

$$\text{s.t.} \quad f_t = f_t^0 + q_t, \quad \forall t \in \mathcal{T} \tag{10b}$$

$$f_t \in [\underline{f}, \bar{f}], \quad \forall t \in \mathcal{T} \tag{10c}$$

$$s_t = s_{t-1} + f_t, \quad \forall t \in \mathcal{T} \tag{10d}$$

$$s_t \in [0, \bar{s}], \quad \forall t \in \mathcal{T} \tag{10e}$$

where $f_t^0$ is the initial position, $\bar{f}$ and $\underline{f}$ are injection and withdrawal limits (MW), and $\bar{s}$ is maximum storage capacity (MWh).

## 4.2 Extension to Continuous Intraday Markets

For continuous intraday markets with limit order books, we extend the intrinsic problem to account for:

- Discrete tradable quantities (multiples of minimum trading unit $u$)
- Order book structure with heterogeneous prices
- Trading costs $\nu_{trade}$ (€/MWh)
- Efficiency losses $\eta^+$ (charging) and $\eta^-$ (discharging)
- Battery degradation costs (€/MWh)

For each product $t$, order $i \in \mathcal{O}_t$ has price $P_i$ and quantity $Q_i$. The traded quantity for order $i$ is $q_i = k_i u$ where $k_i \in \mathbb{N}_0$.

## 4.3 Health-Aware MILP Formulation

The intrinsic health-aware problem that incorporates the piecewise linear degradation model is formulated as follows:

### Decision Variables

- $k_i$: Integer multiplier (blocks) for order $i$.
- $p_t^{ch,j}, p_t^{dis,j}$: Charging and discharging power for cycle depth segment $j$ at time $t$.

- $\alpha_t$: Binary variable for operating mode (1 = charge/buy, 0 = discharge/sell).

$$\max_{\substack{\alpha_t, k_i, \\ p_t^{dis,j}, p_t^{ch,j}}} \sum_{t\in\mathcal{T}} \Bigg[ \sum_{i\in\mathcal{O}_t^-} (P_i - \nu) q_i - \sum_{i\in\mathcal{O}_t^+} (P_i + \nu) q_i - \sum_{j=1}^{J} M c_j p_t^{dis,j} \Bigg] \tag{11a}$$

$$\text{s.t.} \quad 0 \le q_i \le Q_i, \quad \forall i \in \mathcal{O}_t, \forall t \in \mathcal{T} \tag{11b}$$

$$q_i = k_i u, \quad \forall i \in \mathcal{O}_t, \forall t \in \mathcal{T} \tag{11c}$$

$$f_t^+ = \sum_{i\in\mathcal{O}_t^+} q_i, \quad \forall t \in \mathcal{T} \tag{11d}$$

$$f_t^- = \sum_{i\in\mathcal{O}_t^-} q_i, \quad \forall t \in \mathcal{T} \tag{11e}$$

$$f_t = f_t^0 + f_t^+ - f_t^-, \quad \forall t \in \mathcal{T} \tag{11f}$$

$$f_t \in [M\underline{f}, M\bar{f}], \quad \forall t \in \mathcal{T} \tag{11g}$$

$$f_t = M(i_t - w_t), \quad \forall t \in \mathcal{T} \tag{11h}$$

$$i_t \in [0, \alpha_t \bar{f}], \quad \forall t \in \mathcal{T} \tag{11i}$$

$$w_t \in [0, -(1-\alpha_t)\underline{f}], \quad \forall t \in \mathcal{T} \tag{11j}$$

$$s_t = s_{t-1} + M(\eta^+ i_t - \frac{1}{\eta^-} w_t), \quad \forall t \in \mathcal{T} \tag{11k}$$

$$s_t \in [E^{min}, E^{max}], \quad \forall t \in \mathcal{T} \tag{11l}$$

$$w_t = \sum_{j=1}^{J} p_t^{dis,j}, \quad \forall t \in \mathcal{T} \tag{11m}$$

$$i_t = \sum_{j=1}^{J} p_t^{ch,j}, \quad \forall t \in \mathcal{T} \tag{11n}$$

$$e_{t,j} - e_{t-1,j} = M(p_t^{ch,j}\eta^+ - p_t^{dis,j}/\eta^-) \tag{11o}$$

$$p_t^{ch,j} \ge 0, \quad p_t^{dis,j} \ge 0, \quad \forall t \in \mathcal{T}, \forall j \tag{11p}$$

$$s_t = \sum_{j=1}^{J} e_{t,j}, \quad \forall t \in \mathcal{T} \tag{11q}$$

$$0 \le e_{t,j} \le \bar{e}_j, \quad \forall j, \forall t \tag{11r}$$

$$e_{0,j} = e_j^0, \quad \forall j \tag{11s}$$

$$\sum_{j=1}^{J} e_{T,j} \ge E^{final} \tag{11t}$$

$$c_j = \frac{RJ}{\eta^- E^{rate}} \left[ \Phi\left(\frac{j}{J}\right) - \Phi\left(\frac{j-1}{J}\right) \right] \tag{11u}$$

$$\Phi(\delta) = (5.24\text{e-}4)\delta^{2.03} \tag{11v}$$

$$\alpha_t \in \{0,1\}, \quad k_i \in \mathbb{N}_0, \quad \nu \in \mathbb{R}_{\ge 0} \tag{11w}$$

$s_0 = E^0 = \sum_{j=1}^{J} e_{0,j}$

The variables $p_t^{dis,j}$ and $p_t^{ch,j}$ represent discharge and charge power for cycle depth segment $j$, and $e_{t,j}$ tracks energy stored in segment $j$. The maximum energy in segment $j$ is $\bar{e}_j = E^{rate}/J$.

# 5 Dynamic Programming Formulation

## 5.1 Approach Overview

To enable real-time decision-making in continuous intraday markets while accounting for battery degradation, we develop a dynamic programming (DP) approximation of the health-aware MILP. The DP formulation achieves computational tractability by discretizing the state space and approximating the segment-based degradation model while preserving the essential economic and physical constraints.

## 5.2 State Space Definition

Let the battery's state of charge (SoC) be discretized on a uniform grid $G = \{s_1, s_2, \ldots, s_m\}$, where $s_i = E^{\min} + (i-1)\Delta s$ for $i = 1, \ldots, m$, with $\Delta s = (E^{\max} - E^{\min})/(m-1)$. The DP state at time $t$ is denoted by $s_t \in G$.

## 5.3 Action Space Definition

At each decision epoch $t$, the action $a_t$ represents the net energy traded, expressed as an integer multiple $k_t$ of the minimum trading unit $u$:

$$a_t = k_t u, \quad k_t \in \mathbb{Z}$$

Positive $a_t$ indicates charging (buying), negative $a_t$ indicates discharging (selling), and $a_t = 0$ represents idling.

## 5.4 State Transition Dynamics

The battery's state evolves according to:

$$s_{t+1} = s_t + \Delta s(a_t, s_t)$$

where the state transition function $\Delta s(\cdot)$ accounts for efficiency losses:

$$\Delta s(a_t, s_t) = \begin{cases} \eta^{\text{ch}} a_t & \text{if } a_t > 0 \quad \text{(charging)} \\ \dfrac{a_t}{\eta^{\text{dis}}} & \text{if } a_t < 0 \quad \text{(discharging)} \\ 0 & \text{if } a_t = 0 \end{cases}$$

with $\eta^{\text{ch}}, \eta^{\text{dis}} \in (0, 1]$ representing charging and discharging efficiencies.

## 5.5 Trading Profit Function

The immediate trading profit $\pi_t(a_t)$ depends on the limit order book state at time $t$:

$$\pi_t(a_t) = \begin{cases} -\sum_{i \in \mathcal{I}_t^{\text{buy}}(a_t)} (P_i + \nu) q_i & \text{if } a_t > 0 \\ \sum_{i \in \mathcal{I}_t^{\text{sell}}(-a_t)} (P_i - \nu) q_i & \text{if } a_t < 0 \\ 0 & \text{if } a_t = 0 \end{cases}$$

where $\mathcal{I}_t^{\text{buy}}(a_t)$ and $\mathcal{I}_t^{\text{sell}}(a_t)$ are sets of orders executed to fulfill the trade of size $|a_t|$, $P_i$ is the price of order $i$, $q_i$ is the executed quantity, and $\nu = \nu_{\text{trade}}$ incorporates trading fees.

## 5.6 Segment Distribution Approximation

Instead of explicitly tracking energy in each cycle depth segment, we exploit the *optimal filling pattern* property: energy is always stored in the shallowest available segments first during charging and discharged from the shallowest segments first during discharging. This allows us to infer the segment distribution from the total SoC $s_t$.

Let $\Delta e = E^{\text{rate}}/J$ be the energy capacity per segment. The number of completely full segments $n_t$ and the fractional fill $f_t$ of the next segment are:

$$n_t = \left\lfloor \frac{s_t}{\Delta e} \right\rfloor, \quad f_t = \frac{s_t}{\Delta e} - n_t$$

The energy in segment $j$ is then:

$$e_{t,j} = \begin{cases} \Delta e & \text{for } j = 1, \ldots, n_t \\ f_t \Delta e & \text{for } j = n_t + 1 \\ 0 & \text{for } j > n_t + 1 \end{cases}$$

## 5.7 Degradation Cost Model

The degradation cost $C_{\text{deg}}(s_t, a_t)$ is incurred only during discharging ($a_t < 0$). Let $W = |a_t|$ be the discharge energy. We allocate this energy to segments in increasing order of depth $j = 1, 2, \ldots, J$:

**Algorithm 2** Degradation Cost Computation

**Require:** Current SoC $s_t$, discharge energy $W$, segment costs $c_j$, segment width $\Delta e$
**Ensure:** Degradation cost $C_{\text{deg}}$

```
Compute n_t = ⌊s_t/Δe⌋, f_t = s_t/Δe − n_t
Initialize C_deg ← 0, W_rem ← W
for j = 1 to J do
    if j ≤ n_t then
        e_j ← Δe
    else if j = n_t + 1 then
        e_j ← f_t Δe
    else
        e_j ← 0
    end if
    d_j ← min(e_j, W_rem)
    C_deg ← C_deg + c_j · d_j
    W_rem ← W_rem − d_j
    if W_rem = 0 then
        break
    end if
end for=0
```

The segment marginal costs $c_j$ are precomputed from the piecewise linear approximation:

$$c_j = \frac{RJ}{\eta^- E^{\text{rate}}} \left[ \Phi\left(\frac{j}{J}\right) - \Phi\left(\frac{j-1}{J}\right) \right]$$

where $\Phi(\delta) = 5.24 \times 10^{-4} \delta^{2.03}$ is the near quadratic cycle depth stress function for Li(NiMnCo)O2-based 18650 lithium-ion battery cells.

## 5.8 Value Function and Bellman Equation

Let $V_t(s_t)$ denote the maximum expected profit from time $t$ to the end of the trading horizon $T$, starting from state $s_t$. The value function satisfies the Bellman equation:

$$V_t(s_t) = \max_{a_t \in \mathcal{A}_t(s_t)} \left\{ \pi_t(a_t) - C_{\text{deg}}(s_t, a_t) + \hat{V}_{t+1}(s_{t+1}) \right\}$$

where $\mathcal{A}_t(s_t)$ is the set of feasible actions at state $s_t$, and $\hat{V}_{t+1}$ is the interpolated value function at the next state $s_{t+1} = s_t + \Delta s(a_t, s_t)$.

## 5.9 Feasible Action Set

An action $a_t = k_t u$ is feasible if it satisfies:

1. **Power capacity constraints:**

$$\begin{cases} 0 \le a_t \le M \cdot G & \text{if charging} \\ M \cdot D \le a_t \le 0 & \text{if discharging} \end{cases}$$

where $G$ and $D$ are maximum charging and discharging power ratings, and $M$ is the duration of the trading interval.

2. **Energy capacity constraints:**

$$E^{\min} \leq s_t + \Delta s(a_t, s_t) \leq E^{\max}$$

3. **Order book liquidity constraints:**

$$|a_t| \leq \min \left( \sum_{i \in \mathcal{O}_t^{\text{buy}}} Q_i, \sum_{i \in \mathcal{O}_t^{\text{sell}}} Q_i \right)$$

where $Q_i$ is the available volume at the best prices.

## 5.10 Boundary Conditions

The DP terminates at the end of the trading horizon with:

$$V_{T+1}(s) = 0 \quad \forall s \in G$$

A final SoC requirement is enforced during the forward pass:

$$s_T \geq E^{\text{final}}$$

## 5.11 Value Function Interpolation

Since $s_{t+1}$ may not fall exactly on the discretization grid $G$, we use linear interpolation:

$$\hat{V}_{t+1}(s) = \frac{s_{i+1} - s}{\Delta s} V_{t+1}(s_i) + \frac{s - s_i}{\Delta s} V_{t+1}(s_{i+1})$$

for $s_i \leq s \leq s_{i+1}$, where $s_i, s_{i+1} \in G$ are adjacent grid points.

## 5.12 Complete DP Algorithm

**Algorithm 3** Health-Aware Rolling Intrinsic DP

**Require:** State grid $G$, initial SoC $s_0$, order book data $\{\mathcal{O}_t\}_{t=1}^T$, battery parameters
**Ensure:** Optimal action sequence $\{a_t^*\}_{t=1}^T$, total profit $V_1(s_0)$

```
0: Initialization:
0: Precompute segment costs c_j for j = 1, ..., J
0: Precompute degradation costs C_deg(s, a) for all s ∈ G and feasible a
0: Set V_{T+1}(s) ← 0 for all s ∈ G
0: Backward pass:
0: for t = T, T − 1, ..., 1 do
0:   for each s ∈ G do
0:     V_t(s) ← −∞
0:     for each feasible action a at state s do
0:       Compute s' ← s + Δs(a, s)
0:       Compute π ← π_t(a) using order book O_t
0:       C ← C_deg(s, a)
0:       V_next ← interpolate(V_{t+1}, s')
0:       V_total ← π − C + V_next
0:       if V_total > V_t(s) then
0:         V_t(s) ← V_total
0:         policy_t(s) ← a
0:       end if
0:     end for
0:   end for
0: end for
0: Forward pass:
0: s_1 ← s_0
0: for t = 1, 2, ..., T do
0:   a*_t ← policy_t(s_t)
0:   Execute trade a*_t in simulated market
0:   s_{t+1} ← s_t + Δs(a*_t, s_t)
0: end for
0: Return: {a*_t}_{t=1}^T, V_1(s_0) =0
```

## 5.13 Computational Complexity Analysis

The computational complexity of Algorithm 3 is:

$$\mathcal{O}\left(T \cdot |G| \cdot |\mathcal{A}|\right)$$

where:

- $T$ is the number of time periods,
- $|G| = m$ is the number of discretized states,
- $|\mathcal{A}|$ is the average size of the feasible action set.

For typical parameter values ($T \approx 100$, $m \approx 100$, $|\mathcal{A}| \approx 50$), the DP requires approximately $5 \times 10^5$

operations, which can be executed in milliseconds on modern hardware. This represents a speedup of 2–3 orders of magnitude compared to solving the MILP at each time step.

### 5.14 Implementation Optimizations

To further enhance computational efficiency:

1. **Parallelization:** The backward pass computations for different states $s \in G$ are independent and can be parallelized.

2. **Warm-starting:** The value function $V_t(s)$ from the previous optimization can be used as an initial guess for the current optimization, reducing the number of iterations.

3. **Adaptive discretization:** The grid resolution $\Delta s$ can be adapted based on the curvature of the value function, using finer grids where $V_t(s)$ changes rapidly.

4. **Action pruning:** Dominated actions (those with lower profit and higher degradation cost than another available action) can be eliminated from $\mathcal{A}_t(s_t)$.

### 5.15 Convergence Properties

The DP algorithm converges to the optimal solution of the discretized problem as $m \to \infty$ (finer state discretization) and $\Delta s \to 0$. The error introduced by linear interpolation is bounded by $O(\Delta s^2)$ for sufficiently smooth value functions.

### 5.16 Relationship to Original MILP

The DP formulation approximates the original health-aware MILP through:

- State space discretization instead of continuous optimization,

- Implicit segment tracking via the optimal filling pattern,

- Sequential decision-making rather than simultaneous optimization over all time periods.

The approximation error decreases with finer state discretization and more segments $J$ in the degradation model.

## 6 Implementation Framework

### 6.1 The Rolling Intrinsic Policy

The Rolling Intrinsic (RI) policy operates by repeatedly solving the health-aware intrinsic optimization as market conditions evolve, enabling continuous adaptation to new price information and order book updates. This approach is particularly well-suited for continuous intraday markets where price signals change rapidly.

**Algorithm 4** Health-Aware Rolling Intrinsic

**Require:** Initial storage level $s_0$, simulation period $[t_{\text{start}}, t_{\text{end}}]$, LOB data $\mathcal{T}_{t^*}$ for all $t^* \in [t_{\text{start}}, t_{\text{end}}]$, battery parameters $(\bar{f}, \underline{f}, \bar{s}, \eta^{\text{ch}}, \eta^{\text{dis}}, \nu)$

**Ensure:** Final battery schedule, cumulative trading profits

```
for each time instant t* ∈ [t_start, t_end] do
    Solve health-aware intrinsic optimization using DP (Algorithm 3)
    Update all market positions f_t for t ∈ T_{t*}
    Execute trades and update order book
    Log profits/losses from executed trades
    Update battery state s_{t*} based on physical delivery
end for=0
```

The RI policy is fundamentally myopic—it does not explicitly incorporate expectations of future price changes beyond the current trading window. While this characteristic makes it suboptimal in a theoretical sense, it offers several practical advantages:

- **Robustness:** The RI never engages in speculative positions that could result in significant losses. It only enters trades that are immediately profitable given current market conditions and the battery's physical constraints.

- **Computational tractability:** The myopic nature avoids the need for complex stochastic optimization or scenario-based approaches that model uncertain future prices.

- **Adaptability:** By continuously re-solving the optimization problem, the RI can rapidly respond to new information, making it well-suited for high-frequency trading environments.

### 6.2 Parametric Enhancement

The basic RI policy can be enhanced through the introduction of a penalty parameter $\phi$ that discourages trading in illiquid market conditions, characterized by

large bid-ask spreads. This enhancement addresses a key limitation of the myopic approach: the tendency to execute trades too early at unfavorable prices when markets are illiquid.

$$\hat{\pi}_t(a_t) = \pi_t(a_t) - (\phi \cdot \delta_{t^*,t}) \cdot |a_t| \tag{12}$$

where:

- $\hat{\pi}_t(a_t)$ is the adjusted profit for trading action $a_t$ in delivery period $t$
- $\pi_t(a_t)$ is the original profit from equation (12)
- $\phi \geq 0$ is the penalty parameter (unitless)
- $\delta_{t^*,t}$ is the relative bid-ask spread of product $t$ at time $t^*$, defined as:

$$\delta_{t^*,t} = \frac{P_t^{\text{ask}} - P_t^{\text{bid}}}{(P_t^{\text{ask}} + P_t^{\text{bid}})/2}$$

- $|a_t|$ is the absolute traded volume

The parametric enhancement in equation 12 introduces a trade-off between immediacy and execution quality. A larger $\phi$ value encourages the RI to wait for more liquid market conditions (typically closer to delivery time), potentially achieving better prices at the cost of reduced trading flexibility.

#### 6.2.1 Parameter Optimization

The optimal value of $\phi$ can be determined through data-driven optimization using historical market data. We employ a sliding window approach:

1. Divide the historical data into monthly periods: $M_1, M_2, \ldots, M_{12}$
2. For each month $m \geq 2$, use data from the previous month $M_{m-1}$ as a training set
3. Solve the optimization problem:

$$\phi_m^* =_{\phi \geq 0} \text{Profit}_{M_{m-1}}(\phi)$$

   where $\text{Profit}_{M_{m-1}}(\phi)$ is the total profit achieved by the RI with parameter $\phi$ on month $M_{m-1}$
4. Apply $\phi_m^*$ to the current month $M_m$ for out-of-sample evaluation

Given the computational efficiency of our DP implementation, this parameter optimization can be performed quickly. The optimization over $\phi$ is a one-dimensional search that can be efficiently solved using methods such as Brent's algorithm or golden-section search.

#### 6.2.2 Intuition and Implementation

The penalty term $(\phi \cdot \delta_{t^*,t}) \cdot |a_t|$ serves as an opportunity cost for trading in illiquid markets. When the bid-ask spread $\delta_{t^*,t}$ is large, the penalty increases proportionally to the trade size $|a_t|$, discouraging the RI from executing large trades under unfavorable conditions. This mechanism effectively teaches the RI to be more patient, waiting for market liquidity to improve before committing to substantial position changes.

In the DP implementation, this adjustment modifies the immediate profit calculation in the Bellman equation:

$$V_t(s_t) = \max_{a_t \in \mathcal{A}_t(s_t)} \left[ \hat{\pi}_t(a_t) - C_{\text{deg}}(s_t, a_t) + \hat{V}_{t+1}(s_{t+1}) \right]$$

The parametric enhancement represents a form of regularization that balances exploitation of current arbitrage opportunities against the potential for better execution in the near future.

### 6.3 Computational Implementation Details

#### 6.3.1 Order Book Integration

The RI policy interacts with the continuous intraday market through the limit order book (LOB). At each decision epoch $t^*$, the algorithm:

1. **Reads current LOB state:** Extracts best bid/ask prices and available volumes for all tradable products $\mathcal{T}_{t^*}$
2. **Solves the DP:** Computes optimal actions using the current battery state and LOB information
3. **Submits orders:** Places limit orders at the exchange, typically using "all-or-none" order types to ensure physical feasibility
4. **Updates state:** Modifies the battery's state based on executed trades

#### 6.3.2 Handling Market Feedback

Our implementation accounts for the market impact of the algorithm's own orders through:

- **Realistic backtesting:** When an order is placed, it is immediately matched against the historical order book if possible, and the LOB is updated accordingly

- **Order expiration:** Unmatched orders are canceled after a short duration to prevent accumulation of unrealistic positions
- **Physical feasibility:** After each trading decision, the algorithm verifies that the resulting battery schedule remains physically feasible

#### 6.3.3 Performance Metrics

We evaluate the RI policy using several key metrics:

$$\text{Annual Profit} = \sum_{t=1}^{T} (\pi_t(a_t^*) - C_{\text{deg}}(s_t, a_t^*)) \tag{13}$$

$$\text{Battery Cycles per Day} = \frac{1}{365} \sum_{t=1}^{T} \frac{|a_t^*|}{E^{\text{rate}}} \tag{14}$$

$$\text{Sharpe Ratio} = \frac{\mathbb{E}[R] - r_f}{\sigma[R]} \tag{15}$$

$$\text{Maximum Drawdown} = \max_{t \in [0,T]} \left( \max_{u \in [0,t]} P_u - P_t \right) \tag{16}$$

where $R$ represents daily returns, $r_f$ is the risk-free rate, and $P_t$ is the cumulative profit at time $t$.

### 6.4 Comparison with Alternative Strategies

The health-aware RI policy offers distinct advantages over simpler trading approaches:

Table 1: Comparison of Battery Trading Strategies

| Strategy | Degradation Awareness | Computational Speed | Market Adaptation | Profit Po |
|---|---|---|---|---|
| **Simple Arbitrage** | | Fast | Poor | Low |
| **Myopic MILP** | | Slow | Good | Mediu |
| **Full Stochastic** | | Very Slow | Excellent | High |
| **Health-Aware RI (Ours)** | | Fast | Good | Medium- |

The health-aware RI strikes an effective balance between computational efficiency, market responsiveness, and battery health preservation, making it particularly suitable for high-frequency intraday trading.

## 7 Numerical Case Study

### 7.1 Simulation Setup

We evaluate our health-aware RI policy using historical 2021 German continuous intraday market data from EPEX SPOT, focusing on hourly products. The dataset contains order-by-order LOB updates spanning the entire year.

**Battery Parameters:**

- Power rating: $\bar{f} = -\underline{f} = 10$ MW
- Energy capacity: $\bar{s} = 10$ MWh (1-hour duration)
- Round-trip efficiency: $\eta^+ = \eta^- = 0.95$ ($\eta^+\eta^- = 0.9025$)
- Battery technology: Li(NiMnCo)$O_2$-based lithium-ion cells
- Cycle life: 3000 cycles at 80% DoD
- Shelf life: 10 years at 25°C
- Replacement cost: $R = 300,000$ €/MWh
- Stress function: $\Phi(\delta) = (5.24 \times 10^{-4})\delta^{2.03}$

**Market Parameters:**

- Trading fee: $\nu_{trade} = 0.09$ €/MWh (EPEX volume-based fee)
- Degradation cost: $\nu_{deg} = 4.00$ €/MWh
- Total variable cost: $\nu = 4.09$ €/MWh
- Minimum trading unit: $u = 0.1$ MWh
- Price increment: 0.01 €/MWh

**Optimization Parameters:**

- Action discretization: $v = 1$ (finest granularity)
- Storage state grid: $m \in \{11, 51, 101\}$ equidistant points
- Technical delay: 200 ms (including solve time and communication latency)

The degradation cost of $\nu_{deg} = 4$ €/MWh is motivated by projected 2030 battery replacement costs of 60k €/MWh, a 10-year warranty, and approximately 2 cycles per day.

### 7.2 Comparison: DP vs. MILP

We first compare our DP formulation against the exact MILP solution to validate computational efficiency and solution quality. Table **??** shows results for representative weeks in 2021.

**Key Findings:**

- The DP method achieves **100-1000× speedup** compared to MILP

- Even the coarsest grid (DP-11) maintains **competitive solution quality** (typically within 5% of MILP)
- Finer grids (DP-101) achieve near-identical rewards to MILP with 15-130× speedup
- The number of intrinsic solves is nearly identical across methods, but solve time per instance varies dramatically

The superior speed of DP enables true high-frequency trading, solving the intrinsic at every relevant LOB update (approximately 4.6 solves per millisecond in our full-year simulation).

## 7.3 Impact of Trading Frequency

Table **??** shows how trading frequency affects annual profitability for 2021.

**Key Insights:**

- Trading frequency has a **dramatic impact on profitability**: the fastest strategy earns 58% more than hourly optimization
- Even minute-by-minute optimization leaves 14% of potential profits unrealized
- Diminishing returns set in at very high frequencies, but significant gains persist even at millisecond resolution
- The computational efficiency of our DP enables realistic backtesting over a full year with 24 million intrinsic solves in 86 minutes

## 7.4 Parameter Sensitivity Analysis

We evaluate robustness to key battery and optimization parameters. Table **??** summarizes annual rewards for varying parameters.

**Key Observations:**

1. **Efficiency** has the strongest impact: improving from 90% to 99% round-trip efficiency increases profits by nearly 60%
2. **Degradation costs** significantly affect profitability. Projected decreases in battery costs could increase trading profits by 25%+
3. **Smaller batteries** achieve higher specific revenues (€/MWh capacity) due to diminishing returns on larger capacities
4. **Longer-duration batteries** (2h vs. 1h) show lower specific profits, suggesting 1-hour batteries are better suited for intraday arbitrage
5. **Grid granularity** ($m$) has modest impact, with even $m = 11$ providing good results

## 7.5 Degradation Model Validation

We validate the accuracy of our piecewise linear degradation approximation by comparing predicted aging costs $\hat{C}$ against ex-post rainflow-based benchmark costs $RL$.

Figure **??** shows relative error $\epsilon$ as a function of the number of piecewise segments $J$.

**Results:**

- With $J = 4$ segments, approximation error is $\approx 20\%$
- With $J = 16$ segments, error drops below 2%
- With $J = 40$ segments, error is $< 0.5\%$
- The piecewise model provides an **upper bound** on true degradation cost, ensuring conservative operation

## 7.6 Parametric Enhancement Results

We optimize the bid-ask spread penalty parameter $\phi$ in equation (12) using a sliding monthly window: train on month $m - 1$ to optimize trading in month $m$.

**Training Methodology:**

- Use Brent's method to maximize $f(\phi)$ = monthly trading reward
- Run each evaluation on full month of historical LOB data
- Optimal parameters: $\bar{\phi} = 2.1 \pm 1.6$ (mean ± std across 12 months)

**Out-of-Sample Results (2021):**

- Standard RI ($\phi = 0$): 347k€
- Parametric RI (optimized $\phi$): 376k€
- **Improvement: +8.4%** with no additional risk

The penalty parameter successfully encourages the RI to wait for more liquid markets, avoiding premature trades at unfavorable prices in illiquid conditions.

### 7.7 Battery State-of-Charge Analysis

Figure **??** shows the annual SoC schedule determined by the health-aware RI policy.

**Observations:**

- **Daily periodicity**: Battery charges primarily during early morning (3-6 AM) and afternoon (2-4 PM) when prices are typically low
- **Intermediate SoC states**: Counter to intuition, the battery spends significant time at intermediate SoC levels rather than purely full/empty states
- **Increased activity in Q4**: Trading volume increases notably in October-December 2021, correlating with elevated gas prices and market volatility
- **Complex trading patterns**: The LOB-based approach induces non-trivial SoC trajectories that differ from simplified bang-bang strategies

### 7.8 Lifecycle Economics

Table **??** compares lifecycle metrics between degradation-aware and degradation-agnostic strategies.

**Key Findings:**

1. The **degradation-agnostic** strategy maximizes short-term revenue but operates at a **massive loss** when accounting for battery replacement costs
2. Both degradation-aware strategies achieve **positive lifecycle profits** by intelligently managing cycle frequency and depth
3. The **health-aware piecewise model** achieves slightly higher profits (+3.1%) than the simple linear model while providing more accurate degradation predictions
4. Battery life expectancy increases from 1.1 years (agnostic) to 8+ years (health-aware), approaching the 10-year shelf life limit
5. The health-aware strategy reduces cycling from 4.8 to 1.5 cycles/day while maintaining strong profitability

## 8 Discussion

### 8.1 Computational Efficiency

Our DP formulation achieves 100-1000× speedup compared to exact MILP solutions through three key innovations:

**1. Decomposition by product:** Treating each delivery hour as a separate DP stage reduces the problem dimension dramatically. While the monolithic MILP has $O(T \cdot |\mathcal{O}|)$ decision variables, each DP stage involves at most $O(m \cdot |\mathcal{O}_t|)$ evaluations.

**2. Discretization:** Restricting actions to multiples of $v \cdot u$ limits the action space to $O((\bar{f} - \underline{f})/u)$ possibilities per stage, making enumeration feasible.

**3. Value function approximation:** Linear interpolation on grid $\mathcal{G}$ enables continuous state spaces while maintaining $O(m)$ storage and computation per stage.

The total computational complexity is $O(T \cdot m \cdot A)$ where $A$ is the number of feasible actions per stage, compared to the exponential complexity of branch-and-bound for MILP.

### 8.2 Trade-offs and Limitations

**Myopia:** The RI policy is fundamentally myopic, not anticipating future price changes. This leads to:

- Premature trading in illiquid markets far from delivery
- Missed opportunities to wait for better prices
- Inefficient use of storage flexibility

However, our parametric enhancement (Section 5.6) partially addresses this by penalizing illiquid trading.

**Perfect information assumption:** Our backtesting assumes perfect knowledge of how our orders would have been matched. In reality, algorithmic trading by multiple participants can affect market dynamics in ways we cannot capture from historical data alone.

**Linear degradation in segments:** While the piecewise approximation is accurate with sufficient segments, it remains an approximation. More sophisticated degradation models incorporating temperature, C-rate, and SoH could further improve accuracy.

**Single-market focus:** We consider only the continuous intraday market. Multi-market strategies coordinating day-ahead, intraday, and reserve markets could achieve higher profits but require more complex optimization.

## 8.3 Practical Implementation Considerations

Deploying our health-aware RI policy in practice requires addressing several operational challenges:

**1. Forecasting:** Our backtesting assumes perfect price forecasts (i.e., we use realized prices). Real deployment requires forecasting models for intraday prices, which could be integrated into a stochastic DP formulation.

**2. Market impact:** Large BESS may have non-negligible market impact, causing prices to move when placing large orders. This could be addressed through order splitting and more sophisticated execution algorithms.

**3. Technical infrastructure:** Achieving sub-second reaction times requires:

- Low-latency market data feeds
- Colocated servers near exchange infrastructure
- Optimized code (e.g., C++ implementation)
- Parallel processing for multiple simultaneous products

**4. Risk management:** While the RI is conservative by nature, additional safeguards may be needed:

- Position limits to prevent excessive exposure
- Maximum loss thresholds
- Emergency protocols for market disruptions

## 8.4 Extensions and Future Research

Several promising directions for future research include:

**1. Stochastic DP:** Incorporate price forecasts and uncertainty to reduce myopia:

$$V_t(s_{t-1}) = \max_{f_t} \mathbb{E}_{\mathcal{P}_t} \left[\pi_t(f_t) + V_{t+1}(S(s_{t-1}, f_t)) \mid \mathcal{F}_{t^*}\right] \tag{17}$$

where $\mathcal{P}_t$ is the price distribution and $\mathcal{F}_{t^*}$ is the information available at time $t^*$.

**2. Multi-market optimization:** Coordinate bidding across day-ahead, intraday, and reserve markets:

- Reserve provision generates revenue without cycling (pure profit)
- Day-ahead positions can be balanced in intraday
- Requires modeling the temporal sequence and interdependencies

**3. Advanced degradation models:** Incorporate:

- Temperature-dependent aging: $\Phi(\delta, T) = f(T) \cdot \Phi(\delta)$
- C-rate effects: $\Phi(\delta, C) = g(C) \cdot \Phi(\delta)$
- State-of-health tracking: Update $R$ and $\Phi$ based on current SoH

**4. Machine learning enhancements:**

- Deep reinforcement learning for non-myopic policies
- Neural network approximations of value functions
- Learned execution strategies for order placement

**5. Adaptations for emerging markets:**

- Indian Energy Exchange (IEX) with different market rules
- Integration with ancillary service markets
- Vehicle-to-grid (V2G) applications with fleet aggregation

## 8.5 Policy Implications

Our results have important implications for market design and battery deployment:

**1. Value of high-frequency trading:** The 58% profit increase from high-frequency trading suggests that market designs should facilitate algorithmic participation while ensuring fair access for all participants.

**2. Importance of degradation modeling:** The dramatic difference between degradation-aware and agnostic strategies (profitable vs. loss-making) highlights the need for operators to use sophisticated lifecycle economic models.

**3. Battery sizing:** Smaller batteries (5-10 MWh) achieve higher specific revenues than larger ones, suggesting that distributed deployment may be more economical than large centralized installations for arbitrage applications.

**4. Market liquidity:** Our parametric enhancement shows that trading in illiquid markets (large bid-ask spreads) reduces profitability, suggesting that policies to improve market liquidity would benefit storage operators.

# 9 Conclusion

This paper presents a novel health-aware high-frequency trading framework for Battery Energy Storage Systems participating in continuous intraday electricity markets. By integrating the sophisticated piecewise linear battery degradation model of Xu et al. [5] with the efficient Dynamic Programming formulation inspired by Schaurecker et al. [9], we achieve the best of both worlds: computational speed sufficient for millisecond-pace trading and physical realism that ensures long-term asset value preservation.

Our key contributions include:

1. A **health-aware Rolling Intrinsic policy** that accounts for non-linear cycle-depth-dependent battery degradation through a piecewise linear approximation with $J$ segments, achieving $< 2\%$ error with $J = 16$ segments.

2. An **efficient Dynamic Programming formulation** that solves the intrinsic optimization 100-1000× faster than exact MILP methods, enabling reaction to every relevant Limit Order Book update.

3. **Comprehensive backtesting** on 2021 German intraday market data demonstrating that:

   - High-frequency trading (every LOB update) earns 58% more than hourly optimization
   - Health-aware strategies extend battery life from 1.1 to 8+ years
   - Degradation-agnostic strategies operate at massive losses despite high revenues
   - Parametric enhancements can increase profits by an additional 8.4%

4. **Practical insights** on optimal battery sizing, the impact of efficiency and degradation costs, and the trade-offs between trading frequency and computational burden.

Our results demonstrate that truly high-frequency trading is not only feasible but essential for maximizing BESS profitability in continuous intraday markets. However, short-term profit maximization must be balanced against long-term asset health. The health-aware framework we propose achieves this balance, enabling batteries to capture arbitrage opportunities while preserving their lifecycle economic value.

As battery costs continue to decline and renewable penetration increases, the deployment of grid-scale BESS will accelerate. Our framework provides a practical, computationally efficient, and physically realistic approach to operating these valuable assets in fast-moving electricity markets. The open-source implementation we provide (BitePy Python package) will enable researchers and practitioners to build upon this work and develop even more sophisticated trading strategies.

Future research should focus on incorporating price forecasts to reduce myopia, coordinating across multiple markets (day-ahead, intraday, reserves), and adapting the framework to emerging markets such as the Indian Energy Exchange. By continuing to bridge the gap between computational efficiency and physical realism, we can unlock the full potential of battery storage to enable the renewable energy transition.

# Acknowledgments

The author thanks Dr. Debaprasad Sahu for supervision and guidance, and acknowledges the valuable contributions of the research papers by Xu et al. [5] and Schaurecker et al. [9] which form the theoretical foundation of this work.

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